Year 10
Pathway 10A

Mini-Test 4
Probability

Baldivis
Secondary College

Name: Solutions   Class: ______   Time: 30 minutes   Total Mark:____/26

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning will be allocated zero marks.

Resources: 1 A4 page of notes (1 side), Calculator allowed.

Question 1	**5 marks (1 ,1, 1, 1,1)**

Consider rolling a red and a black die and the probabilities of the following events:

Event A   the red die lands on 5 $\frac{1}{6}$

Event B   the black die lands on 2 $1/6$.

Event C   the sum of the dice is 10. $\cancel{3/36}$ $\frac{1}{12}$.

a) The initial probability of each event described is: (Circle the correct answer)

A $P(A) = \frac{1}{6}$, $P(B) = \frac{1}{6}$, $P(C) = \frac{1}{6}$

B $P(A) = \frac{5}{6}$, $P(B) = \frac{2}{6}$, $P(C) = \frac{7}{36}$

C $P(A) = \frac{5}{6}$, $P(B) = \frac{2}{6}$, $P(C) = \frac{5}{18}$

D $P(A) = \frac{1}{6}$, $P(B) = \frac{1}{6}$, $P(C) = \frac{1}{12}$

E $P(A) = \frac{1}{6}$, $P(B) = \frac{2}{6}$, $P(C) = \frac{1}{12}$

b) Calculate the following probabilities:

i) $P(A|B)$

$\frac{1}{6}$

ii) $P(A \cap B)$

$\cancel{\frac{1}{36}}$ $\frac{1}{36}$?

iii) $P(C|A)$

$\frac{1}{6}$

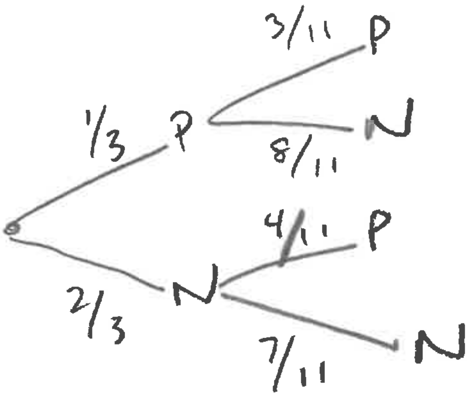
iv) $P(A \cap B \mid C)$

0.

Question 2	**8 marks (4, 1, 1, 2)**

Sam has 12 cards numbered 1 to 12. She draws 2 cards, one after another, without placing them back.
Let P= {number divisible by 3} and N= {number not divisible by 3}

a. Draw a tree diagram to show this information



$\text{Prob}(P,P) = \frac{1}{3} \times \frac{3}{11} = \frac{3}{33} = \frac{1}{11}$

$\text{Prob}(P,N) = \frac{1}{3} \times \frac{8}{11} = \frac{8}{33}$

$\text{Prob}(N,P) = \frac{2}{3} \times \frac{4}{11} = \frac{8}{33}$

$\text{Prob}(N,N) = \frac{14}{33}$

b. Find P (both cards show a number divisible by 3)

$P(P,P) = \frac{1}{11}$

c. Find P (no cards show a number divisible by 3)

$P(N,N) = \frac{14}{33}$

d. Find P(at least one card shows a number divisible by 3)

$P(P,P) + P(P,N) + P(N,P)$

$= \frac{19}{33}$

$(\text{or } 1 - P(N,N))$

Question 3	7 marks (3, 1, 1, 2)

From the 20 people in an aquatic team, 15 like swimming (S), eight like diving (D) and three like both swimming and diving.

a. Represent the information in a two-way table.

	S	S'	T
D	3	5	8
D'	12	0	12
T	15	5	20

b. A person from the aquatic team is chosen at random. Find the probability that the person:

i likes both swimming and diving.

$Pr(S \cap D) = 3/20$

ii likes only diving.

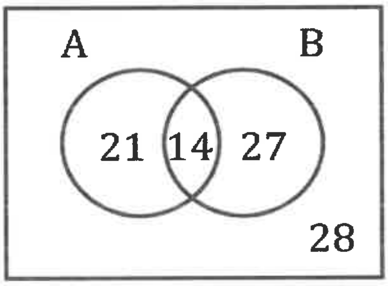
$Pr(D \cap S') = 5/20 = 1/4$

iii likes swimming given that they like diving.

$P(S | D) = 3/8$

Question 4	**7 marks (1, 1, 1, 1, 1, 2)**

Consider the following Venn Diagram:



a Find:

i $n(A')$ $= 27 + 28 = 55$

ii $n(A' \cup B')$ $= 21 + 27 + 28 = 76$

b Find:

i Pr(A) $Pr(A) = 35/90 = 7/18$

ii $Pr(A \cup B')$ $= 28 + 21 + 14 = 63/90 = 7/10$

iii $Pr(B \mid A)$ $= \dfrac{Pr(B \cap A)}{Pr(A)} = \dfrac{14}{35} = \dfrac{2}{5}$

c Are the events A and B independent? Why/why not?

Not equal, not independent

or

$Pr(A \mid B) \neq Pr(A)$ (or something similar)

END OF TEST